
What Counts as a Mistake? Annotating Recitation Events in Quran Memorization Transcripts

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Abstract

1 An application that checks Quran recitation has to separate a genuine error from
2 the ordinary business of reciting from memory: catching a breath and repeating
3 three words, restarting an ayah, correcting oneself mid-word, opening with the
4 isti'adhah. We report a completed human annotation of 100 production recordings
5 under a taxonomy that makes that separation explicit, covering 348 scored units
6 and 162 localized events across ten combined labels, and we freeze an executable
7 scorecard for it. Two existing systems, a plain diff and the segmentation-and-repair
8 pipeline of a deployed application, reach a localization F1 of 0.93 and 0.79 while
9 reaching a label-aware F1 of only 0.53 and 0.45. Both score zero on eight of the ten
10 labels, and the diff calls the opening formulas errors 46 times. The open problem
11 in recitation checking is not finding textual differences but naming them.

1 The distinction the task turns on

13 Quran memorization is normally practised with a listener who corrects the reciter. An application
14 that stands in for that listener transcribes with commercial ASR and then compares the transcript
15 against the reference. Edit distance answers a question nobody asked. In our production data 74% of
16 reviewed ayah units carry no event at all, and among those that do, a substantial share are things a
17 human listener would never mention: the reciter repeated a phrase after a breath, said a word wrongly
18 and immediately fixed it, or used a written form differing from the Uthmani orthography without
19 differing in what was recited. An application that reports these loses the learner's trust faster than one
20 that occasionally misses a real slip.

21 We therefore annotate transcript-against-reference differences as typed, localized *events*, and we
22 judge the text rather than the speaker: a label records what differs between transcript and reference,
23 not whether the reciter or the ASR caused it. Unwritten vowel and tajweed errors lie outside a
24 transcript-only rubric and outside this work.

2 Taxonomy and annotation

26 **Combined labels.** An earlier pilot used two parallel fields, an event type and a be-
27 nign/mistake verdict, plus an uncertain escape hatch. Reviewing real recordings made
28 both untenable: the verdict was never independent of the type in practice, and uncertain
29 absorbed exactly the cases that needed a decision. The rubric now uses one com-
30 bined label per event, `substitution_mistake`, `omission_mistake`, `insertion_mistake`,
31 `substitution_corrected`, `omission_corrected`, `repetition_benign`, `letters_benign`,
32 `spelling_benign`, `basmala_benign` and `isti3adha_benign`, with `clean` for a reviewed unit

33 carrying no event. A unit reviewed and found clean is not the same as a unit not yet reviewed, and the
34 data distinguishes them.

35 **Four decisions that took adjudication.** *Omissions* include missing beginnings, interiors and endings,
36 with no exemption for the final ayah of a recording: stopping early still leaves material the reciter did
37 not produce, and a reviewer cannot infer intent from recording position. *Repeats* span the original
38 phrase together with its extra occurrence against a single reference copy, aligning their localization
39 with corrected events; a wrong-then-correct attempt is a corrected event, while a correct-then-incorrect
40 restatement is a *substitution_mistake* spanning both attempts, with the order recorded in the
41 note. *Spelling* differences removed by an agreed context-independent normalization receive no event,
42 and only a residual form needing contextual acceptance receives *spelling_benign*; the accepted
43 contexts are enumerated rather than generalized, so the category cannot quietly become blanket final-
44 letter deletion. *Localization* uses verbatim words and half-open spans into the original whitespace
45 token arrays, with an empty hypothesis span anchoring an omission and an empty reference span
46 anchoring an insertion.

47 **Scale and composition.** All 100 selected recording cases are human-approved: 314 ayah units,
48 233 of them clean, 115 within-ayah events and 47 opening-formula annotations, over 274 ayahs
49 in 38 surahs. The event distribution is uneven by nature, from 62 *substitution_mistake* down
50 to one *insertion_mistake* and one *omission_corrected*, and any evaluation has to survive
51 that. Ten learner identities contribute, one of them 46 recordings. Selection was purposive, so these
52 counts describe the annotated set; they do not estimate error prevalence or imply generalization to
53 unseen reciters. Three further approved recordings are held as reserves outside the scored set. We
54 verified identifiers and transcripts against both source exports and excluded punctuation-normalized
55 duplicates and redundant clean subpassages. A deterministic 30-record sample covering every label,
56 including the singletons, is available to reviewers.

57 3 An executable scorecard

58 Systems receive the reviewed ayah split, the ayah ids and the exact references, so detection and
59 segmentation are supplied rather than scored; that task is treated in a companion submission. Per unit
60 a system returns events of the form {label, hyp_span, ref_span}, with half-open spans into
61 the transcript and reference token arrays.

62 **Opening formulas are scored.** Each record’s isti’adhah and basmala are presented as their own unit,
63 with the formula as the transcript and an empty reference; a compound opening yields two events
64 over one token array. Deciding that a basmala preceding the ayah is benign rather than an insertion is
65 exactly the judgment under test, and excluding it would hide the failure mode that turns out to matter
66 most.

67 **Matching.** Predictions and gold events are paired one-to-one within a unit by maximum total
68 similarity, so a single broad prediction is credited with at most one gold event and the remainder count
69 as misses. A pair is eligible only when *both* spans overlap, $\min(\text{sim}_{hyp}, \text{sim}_{ref}) \geq 0.30$; averaging
70 the two, as an earlier version did, lets an exact reference span carry a match while the hypothesis span
71 points somewhere else entirely. An empty span is an anchor, matching another anchor within one
72 token and a real span when it falls inside it. Pairing ignores labels, which is what allows localization
73 to be reported apart from naming.

74 **Scores.** The primary measure is label-aware event F1: a paired prediction is a true positive only
75 when its label also matches, and a mislabeled pair counts once as a false positive and once as a
76 false negative. We report micro F1 as the headline, macro F1 so the rare labels cannot be ignored,
77 and localization F1 from the same matching with labels dropped. An application-facing cost is
78 reported separately in raw event counts, (2 missed + false flags) per 100 units, so that the stated
79 exchange rate of one missed mistake for two unnecessary flags is the one the formula implements.
80 Predicting nothing gives micro F1 0; the gold annotation gives 1. Seven oracle tests run against the
81 real annotation before any system is scored, including four that an earlier scorecard passed perfectly
82 while being wrong: mislabeling every mistake, omitting every benign and corrected event, fabricating
83 benign events on clean units, and localizing every event to the wrong place.

System	micro F1	macro F1	loc F1	span IoU	review cost
Predict nothing	0.000	0.000	0.000	—	51.2
Plain diff	0.531	0.179	0.931	0.893	18.1
Deployed application pipeline	0.447	0.170	0.786	0.883	26.7

Table 1: 348 scored units, 162 gold events. Both systems locate events well and name them poorly.

84 4 What existing systems do

85 The gap between the localization and label columns is the finding. A plain diff pairs 93% of the
86 gold events and gets barely half of their labels right. The deployed pipeline, which already models
87 repetition and restart explicitly, does worse on both, because its repair step deletes the material an
88 annotator marks: cleaning the transcript before comparing destroys the evidence for the benign event.

89 The per-label breakdown is starker than the aggregate. The diff reaches F1 0.94 on
90 omission_mistake and 0.85 on substitution_mistake and scores exactly zero on the other
91 eight labels. It emits 46 insertion_mistake predictions where the annotation contains one,
92 because it reads the isti’adhah and the basmala as inserted text. Neither system ever proposes
93 repetition_benign, spelling_benign, letters_benign or either corrected label. Macro F1
94 near 0.17 is what two working categories out of ten looks like.

95 This is a useful negative result for anyone building such an application. Engineering effort in a
96 recitation checker typically goes into alignment quality, and alignment quality is not the binding
97 constraint: both systems already find the differences. What neither does is decide what a difference
98 means, and that decision is what the learner experiences.

99 5 Limitations and what comes next

100 The annotated set is purposive and small, one learner contributes 46 of the 100 recordings, and the
101 label distribution is heavily skewed, so macro F1 here is a coverage check rather than a stable estimate.
102 More consequentially, all 100 selected transcripts already appear as inputs in a public release of
103 the same recordings, and twelve appear in an earlier machine-labeled pilot. The adjudications and
104 the selection membership are private, but the inputs are not, and runtime isolation cannot establish
105 absence of prior exposure. Any claim about unseen inputs requires a separately collected corpus; a
106 read-only production check has since identified 63 further recordings, 57 textually distinct from the
107 released export, as the start of one, subject to reuse permission and de-identification.

108 The annotation and the scorecard exist to support a measured experiment we have not yet run: supply
109 an agent with an unlabeled development pool, the reference and the label definitions, let it produce
110 its own annotations and an algorithm, and score only the resulting algorithm against the private set.
111 Agreement with self-generated labels would not be independent annotation accuracy, and the score
112 assesses the combined process rather than any single annotation’s contribution. Budgets, comparison
113 conditions and permitted feedback remain to be frozen. Gold answers, selection membership, reviewer
114 files and annotation-session memory stay outside every agent runtime, and shared documentation
115 carries aggregate findings and synthetic examples only.